

# Лабораторная работа №6

Установка и настройка системы управления базами данных MariaDB

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## 1. 1. Цели и задачи работы

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## 1.1 Цель лабораторной работы

Получение практических навыков установки и настройки системы управления базами данных MariaDB, настройки кодировки UTF-8, создания и администрирования базы данных, резервного копирования и восстановления данных, а также автоматизации конфигурации MariaDB средствами Vagrant.

## 2. 2. Процесс выполнения лабораторной работы

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## 2.1 Устанавливаю MariaDB и просматриваю конфигурацию

```
Installed:
 mariadb-3:10.11.18-1.el10_2.x86_64
 mariadb-client-utils-3:10.11.18-1.el10_2.x86_64
 mariadb-errmsg-3:10.11.18-1.el10_2.noarch
 mariadb-server-3:10.11.18-1.el10_2.x86_64
 mysql-selinux-1.0.14-1.el10_0.noarch
 perl-Sys-Hostname-1.25-515.el10_2.x86_64
 mariadb-backup-3:10.11.18-1.el10_2.x86_64
 mariadb-common-3:10.11.18-1.el10_2.noarch
 mariadb-gssapi-server-3:10.11.18-1.el10_2.x86_64
 mariadb-server-utils-3:10.11.18-1.el10_2.x86_64
 perl-DBD-MariaDB-1.23-10.el10.x86_64

Complete!
[root@server.paulinelourdes.net ~]# ls /etc/my.cnf.d/
auth_gssapi.cnf  enable_encryption.preset  mysql-clients.cnf  provider_lz4.cnf  provider_snappy.cnf
client.cnf      mariadb-server.cnf        provider_bzip2.cnf  provider_lzo.cnf  spider.cnf
[root@server.paulinelourdes.net ~]# cat /etc/my.cnf
#
# This group is read both both by the client and the server
# use it for options that affect everything
#
[client-server]

#
# include all files from the config directory
#
!includedir /etc/my.cnf.d

[root@server.paulinelourdes.net ~]# █
```

Рисунок 1: Установка MariaDB и просмотр конфигурационных файлов

## 2.2 Запускаю MariaDB и проверяю порт

```
[root@server.paulinelourdes.net ~]# systemctl start mariadb
[root@server.paulinelourdes.net ~]# systemctl enable mariadb
Created symlink '/etc/systemd/system/mysql.service' → '/usr/lib/systemd/system/mariadb.service'.
Created symlink '/etc/systemd/system/mysqld.service' → '/usr/lib/systemd/system/mariadb.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/mariadb.service' → '/usr/lib/systemd/system/mariadb.service'.
[root@server.paulinelourdes.net ~]# ss -tulpen | grep mysql
[root@server.paulinelourdes.net ~]# ss -tulpen | grep maria
tcp LISTEN 0      80                      0.0.0.0:3306           0.0.0.0:*             users:((("mariadb",pid=12444,fd=20)
)
                                uid:27 ino:59044 sk:1a cgr
oup:/system.slice/mariadb.service <->
tcp LISTEN 0      80                      [::]:3306             [::]:*                users:((("mariadb",pid=12444,fd=21)
)
                                uid:27 ino:59045 sk:28 cgr
oup:/system.slice/mariadb.service v6only:1 <->
[root@server.paulinelourdes.net ~]#
```

Рисунок 2: Запуск службы MariaDB и проверка прослушивания порта 3306

## 2.3 Настраиваю безопасность MariaDB

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n]

New password:

Re-enter new password:

Password updated successfully!

Reloading privilege tables..

... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone to log into MariaDB without having to have a user account created for them. This is intended only for testing, and to make the installation go a bit smoother. You should remove them before moving into a production environment.

Remove anonymous users? [Y/n]

... Success!

Normally, root should only be allowed to connect from 'localhost'. This ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n]

... Success!

By default, MariaDB comes with a database named 'test' that anyone can access. This is also intended only for testing, and should be removed before moving into a production environment.

Remove test database and access to it? [Y/n]

Рисунок 3: Настройка параметров безопасности MariaDB с помощью mysql\_secure\_installation

## 2.4 Просматриваю команды MariaDB

```
General information about MariaDB can be found at
http://mariadb.org

List of all client commands:
Note that all text commands must be first on line and end with ';'
?          (\?) Synonym for 'help'.
charset    (\C) Switch to another charset. Might be needed for processing binlog with multi-byte charsets.
clear       (\c) Clear the current input statement.
connect     (\r) Reconnect to the server. Optional arguments are db and host.
delimiter   (\d) Set statement delimiter.
edit        (\e) Edit command with $EDITOR.
ego         (\G) Send command to MariaDB server, display result vertically.
exit        (\q) Exit mysql. Same as quit.
go          (\g) Send command to MariaDB server.
help        (\h) Display this help.
nopager     (\n) Disable pager, print to stdout.
notee       (\t) Don't write into outfile.
nowarning   (\w) Don't show warnings after every statement.
pager       (\P) Set PAGER [to_pager]. Print the query results via PAGER.
print       (\p) Print current command.
prompt      (\R) Change your mysql prompt.
quit        (\q) Quit mysql.
rehash      (\#) Rebuild completion hash.
sandbox     (\-) Disallow commands that access the file system (except \P without an argument and \e).
source      (\.) Execute an SQL script file. Takes a file name as an argument.
status      (\s) Get status information from the server.
system      (\!) Execute a system shell command.
tee         (\T) Set outfile [to_outfile]. Append everything into given outfile.
use         (\u) Use another database. Takes database name as argument.
warnings    (\W) Show warnings after every statement.

For server side help, type 'help contents'

MariaDB [(none)]>
```

Рисунок 4: Просмотр списка команд интерактивной оболочки MariaDB

## 2.5 Просматриваю системные базы данных

```
MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database          |
+-----+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+-----+
4 rows in set (0.001 sec)

MariaDB [(none)]> exit;
Bye
[root@server.paulinelourdes.net ~]#
```

Рисунок 5: Просмотр системных баз данных MariaDB

## 2.6 Проверяю исходную кодировку MariaDB

```
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 12
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.11.18-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          12
Current database:
Current user:           root@localhost
SSL:                    Not in use
Current pager:          stdout
Using outfile:          ''
Using delimiter:        ;
Server:                 MariaDB
Server version:         10.11.18-MariaDB MariaDB Server
Protocol version:       10
Connection:             Localhost via UNIX socket
Server characterset:    latin1
Db characterset:        latin1
Client characterset:    utf8mb3
Conn. characterset:     utf8mb3
UNIX socket:            /var/lib/mysql/mysql.sock
Uptime:                 3 min 48 sec

Threads: 1  Questions: 23  Slow queries: 0  Opens: 20  Open tables: 13  Queries per second avg: 0.100
-----

MariaDB [(none)]> █
```

Рисунок 6: Статус MariaDB до изменения кодировки сервера

## 2.7 Настраиваю кодировку UTF-8

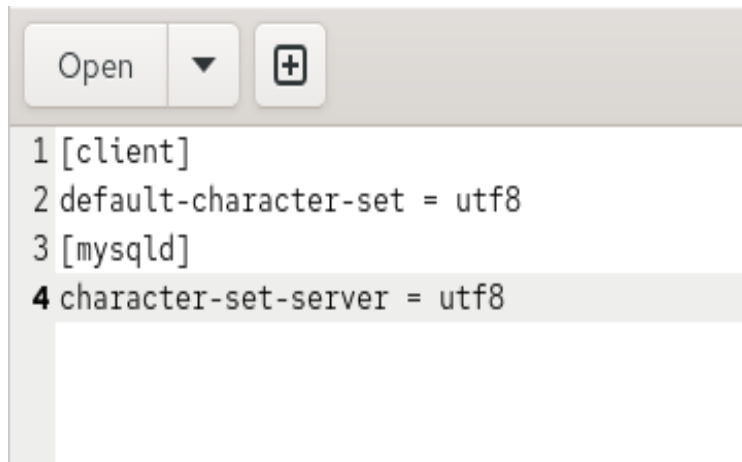


Рисунок 7: Настройка кодировки UTF-8 в файле utf8.cnf

## 2.8 Проверяю изменение кодировки

```
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql  Ver 15.1 Distrib 10.11.18-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          3
Current database:
Current user:           root@localhost
SSL:                    Not in use
Current pager:          stdout
Using outfile:          ''
Using delimiter:        ;
Server:                 MariaDB
Server version:         10.11.18-MariaDB MariaDB Server
Protocol version:       10
Connection:             Localhost via UNIX socket
Server characterset:    utf8mb3
Db characterset:        utf8mb3
Client characterset:    utf8mb3
Conn. characterset:     utf8mb3
UNIX socket:            /var/lib/mysql/mysql.sock
Uptime:                 16 sec

Threads: 1  Questions: 4  Slow queries: 0  Opens: 17  Open tables: 10  Queries per second avg: 0.250
-----
MariaDB [(none)]>
```

Рисунок 8: Статус MariaDB после установки UTF-8 в качестве кодировки по умолчанию

## 2.9 Создаю базу addressbook и таблицу city

```
MariaDB [(none)]>
MariaDB [(none)]> CREATE DATABASE addressbook CHARACTER SET utf8 COLLATE utf8_general_ci;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> USE addressbook;
Database changed
MariaDB [addressbook]> SHOW TABLES;
Empty set (0.000 sec)

MariaDB [addressbook]> CREATE TABLE city (name VARCHAR(40), city VARCHAR(40));
Query OK, 0 rows affected (0.014 sec)

MariaDB [addressbook]> INSERT INTO city(name, city) VALUES ('Иванов','Москва');
Query OK, 1 row affected (0.001 sec)

MariaDB [addressbook]> INSERT INTO city(name, city) VALUES ('Петров','Сочи');
Query OK, 1 row affected (0.001 sec)

MariaDB [addressbook]> INSERT INTO city(name, city) VALUES ('Сидоров','Дубна');
Query OK, 1 row affected (0.003 sec)

MariaDB [addressbook]> SELECT * FROM city;
+-----+-----+
| name          | city          |
+-----+-----+
| Иванов        | Москва       |
| Петров        | Сочи         |
| Сидоров       | Дубна        |
+-----+-----+
3 rows in set (0.000 sec)

MariaDB [addressbook]> █
```

Рисунок 9: Создание базы addressbook, таблицы city и добавление записей

## 2.10 Создаю пользователя и назначаю права

```
MariaDB [addressbook]>
MariaDB [addressbook]> CREATE USER paulinelourdes@'%' IDENTIFIED BY '123456';
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO paulinelourdes@'%';
Query OK, 0 rows affected (0.002 sec)

MariaDB [addressbook]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [addressbook]> DESCRIBE city;
+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| name  | varchar(40)   | YES  |     | NULL    |       |
| city  | varchar(40)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [addressbook]> quit
Bye
[root@server.paulinelourdes.net my.cnf.d]#
```

Рисунок 10: Создание пользователя MariaDB, назначение прав и просмотр структуры таблицы city

## 2.11 Проверяю базу данных с помощью mysqlshow

```
[root@server.paulinelourdes.net my.cnf.d]# mysqlshow -u root -p
Enter password:
+-----+
| Databases |
+-----+
| addressbook |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
[root@server.paulinelourdes.net my.cnf.d]# mysqlshow -u root -p addressbook
Enter password:
Database: addressbook
+-----+
| Tables |
+-----+
| city |
+-----+
[root@server.paulinelourdes.net my.cnf.d]# mysqlshow -u paulinelourdes -p addressbook
Enter password:
Database: addressbook
+-----+
| Tables |
+-----+
| city |
+-----+
[root@server.paulinelourdes.net my.cnf.d]#
```

Рисунок 11: Проверка базы addressbook и таблицы city с помощью mysqlshow

## 2.12 Создаю и восстанавливаю резервные копии

```
[root@server.paulinelourdes.net my.cnf.d]#  
[root@server.paulinelourdes.net my.cnf.d]# mkdir -p /var/backup  
[root@server.paulinelourdes.net my.cnf.d]# mysqldump -u root -p addressbook > /var/backup/addressbook.sql  
Enter password:  
[root@server.paulinelourdes.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > /var/backup/addressbook.sql.gz  
Enter password:  
[root@server.paulinelourdes.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > $(date +%Y  
%m%d.%H%M%S).sql.gz)  
Enter password:  
[root@server.paulinelourdes.net my.cnf.d]# mysql -u root -p addressbook < /var/backup/addressbook.sql  
Enter password:  
[root@server.paulinelourdes.net my.cnf.d]# zcat /var/backup/addressbook.sql.gz | mysql -u root -p addressbook  
Enter password:  
[root@server.paulinelourdes.net my.cnf.d]#
```

**Рисунок 12:** Создание и восстановление резервных копий базы данных addressbook

## 2.13 Автоматизирую настройку MariaDB

```
1  #!/bin/bash
2  echo "Provisioning script $0"
3  systemctl restart named
4  echo "Install needed packages"
5  dnf -y install mariadb mariadb-server
6  echo "Copy configuration files"
7  cp -R /vagrant/provision/server/mysql/etc/* /etc
8  mkdir -p /var/backup
9  cp -R /vagrant/provision/server/mysql/var/backup/* /var/backup
10 echo "Start mysql service"
11 systemctl enable mariadb
12 systemctl start mariadb
13 if [[ ! -d /var/lib/mysql/mysql ]]
14 then
15     echo "Securing mariadb"
16     mysql_secure_installation <<EOF
17     Y
18     123456
19     123456
20     Y
21     Y
22     Y
23     Y
24     EOF
25     echo "Create database"
26     mysql -u root -p123456 <<EOF
27     CREATE DATABASE addressbook CHARACTER SET utf8 COLLATE utf8_general_ci;
28     EOF
29     mysql -u root -p123456 addressbook < /var/backup/addressbook.sql
30 fi
```

Рисунок 13: Скрипт автоматической установки и настройки MariaDB mysql.sh

### 3. 3. Выводы по проделанной работе

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В ходе лабораторной работы была выполнена установка и настройка системы управления базами данных MariaDB на виртуальной машине **server**.

Была настроена безопасность MariaDB, установлена кодировка UTF-8, создана база данных **addressbook** с таблицей **city**, добавлены тестовые записи и настроены права отдельного пользователя.

Также были созданы и проверены резервные копии базы данных, включая сжатый вариант, и выполнено восстановление данных.